

DAY 23 TASK.py - C:/Users/krsuk/OneDrive

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```
# Task 1: String Operations
```

```
text = "DataScienceWithPython"
```

```
print(text[:4])
print(text[-6:])
print(text[::-1])
print(text[::2])
print(text.lower())
```

```
vowels = "aeiouAEIOU"
```

```
count = 0
```

```
for ch in text:
    if ch in vowels:
        count += 1
```

```
print(count)
```

```
===== RESTART: C:/User
```

```
Data
```

```
Python
```

```
nohtyPhtiWecneicSataD
```

```
DtSineihyhn
```

```
datasciencewithpython
```

```
7
```

```
>
```

```
# Task 2: List Operations
```

```
numbers = [10, 20, 30, 40, 50, 20, 10]
```

```
numbers.append(60)
```

```
numbers.insert(1, 15)
```

```
numbers = list(set(numbers))
```

```
numbers.sort(reverse=True)
```

```
print(numbers.count(20))
```

```
numbers_tuple = tuple(numbers)
```

```
print(numbers_tuple)
```

```
===== RESTART: C:/Users/krsuk
```

```
1
```

```
(60, 50, 40, 30, 20, 15, 10)
```

Task 3: Tuple Operations

```
data = (5, 10, 15, 20, 25, 10)
```

```
print(data[3])
```

```
lst = list(data)
```

```
lst.append(30)
```

```
print(lst)
```

```
print(data[1:4])
```

```
===== RESTART: C:/Users/
```

```
20
```

```
[5, 10, 15, 20, 25, 10, 30]
```

```
(10, 15, 20)
```

Task 4: Dictionary Operations

```
student = {  
    "name": "Subrahmanya",  
    "age": 22,  
    "marks": 89  
}
```

```
student["grade"] = "A"
```

```
student["marks"] = 90
```

```
student.pop("age")
```

```
print(student.keys())
```

```
print(student.values())
```

```
print("name" in student)
```

```
===== RESTART: C:/Users/krsu/OneD
dict_keys(['name', 'marks', 'grade'])
dict_values(['Subrahmanya', 90, 'A'])
True
```

Task 5: Set Operations

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

```
print(A | B)
print(A & B)
print(A - B)
print(A ^ B)
```

```
A.add(10)
A.remove(2)
print(A)
```

```
===== RESTART:
{1, 2, 3, 4, 5, 6}
{3, 4}
{1, 2}
{1, 2, 5, 6}
{1, 3, 4, 10}
```

```

# Task 6: Functions

def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, n):
        if n % i == 0:
            return False
    return True

def factorial(n):
    if n == 0:
        return 1
    return n * factorial(n - 1)

def fibonacci(n):
    a, b = 0, 1
    for _ in range(n):
        print(a, end=" ")
        a, b = b, a + b
    print()

def greet(name="User"):
    print("Hello", name)

square = lambda x: x * x

print("Prime check for 7:", is_prime(7))
print("Factorial of 5:", factorial(5))

print("Fibonacci series:")
fibonacci(7)

greet()
greet("Subrahmanya")

print("Square of 6:", square(6))

```

```

===== RESTART: C:/Users/1
Prime check for 7: True
Factorial of 5: 120
Fibonacci series:
0 1 1 2 3 5 8
Hello User
Hello Subrahmanya
Square of 6: 36

```

```
# Task 7: Loops & Logic

def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, n):
        if n % i == 0:
            return False
    return True

num = 5
for i in range(1, 11):
    print(num, "x", i, "=", num * i)

for i in range(1, 6):
    print("*" * i)

n = 123
s = 0
while n > 0:
    s += n % 10
    n //= 10
print(s)

lst = [1, 2, 3, 4, 5, 6]
even = odd = 0
for i in lst:
    if i % 2 == 0:
        even += 1
    else:
        odd += 1
print(even, odd)

count = 0
num = 2
while count < 10:
    if is_prime(num):
        print(num, end=" ")
        count += 1
    num += 1
print()
```

===== RESTART: C:/User

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50

*

**

6

3 3

2 3 5 7 11 13 17 19 23 29

```
# Task 8: OOPS Concepts
```

```
# ----- Class & Object -----
```

```
class Student:
    def __init__(self, name, age, marks):
        self.name = name
        self.age = age
        self.marks = marks

    def display(self):
        print("Name :", self.name)
        print("Age :", self.age)
        print("Marks :", self.marks)
```

```
s1 = Student("Subrahmanya", 22, 86)
s1.display()
```

```
print()
```

```
# ----- Rectangle Class -----
```

```
class Rectangle:
    def __init__(self, length, width):
        self.length = length
        self.width = width

    def area(self):
        return self.length * self.width

    def perimeter(self):
        return 2 * (self.length + self.width)
```

```
a = Rectangle(2, 4)
print("Area:", a.area())
print("Perimeter:", a.perimeter())
```

```
print()
```

```
# ----- Class Variable -----
```

```
class Student:
    school = "Sahyadri"

    def __init__(self, name, age):
        self.name = name
        self.age = age

    def display(self):
        print("School:", Student.school)
        print("Name:", self.name)
```

```
print(school, teacher.school,
      print("Name:", self.name)
      print("Age:", self.age)

s1 = Student("Subrahmanya", 22)
s1.display()

print()
# ----- Inheritance -----
class Animal:
    def sound(self):
        print("Animal makes sound")

class Dog(Animal):
    def bark(self):
        print("Dog barks")

t = Dog()
t.sound()
t.bark()

print()
# ----- Method Overriding -----
class Person:
    def speak(self):
        print("Person speaking")

class Teacher(Person):
    def speak(self):
        print("Teacher speaking")

t = Teacher()
t.speak()

print()
# ----- Encapsulation -----
class Student:
    def __init__(self, marks):
        self._marks = marks

    def show_marks(self):
        print("Marks:", self._marks)

s = Student(90)
s.show_marks()

print()
```



```
print()
# ----- Polymorphism -----
class Dog:
    def sound(self):
        print("Bark")

class Cat:
    def sound(self):
        print("Meow")

def make_sound(animal):
    animal.sound()

d = Dog()
c = Cat()
make_sound(d)
make_sound(c)
```

===== RESTART: C:/Users/kr

Name : Subrahmanya

Age : 22

Marks : 86

Area: 8

Perimeter: 12

School: Sahyadri

Name: Subrahmanya

Age: 22

Animal makes sound

Dog barks

Teacher speaking

Marks: 90

Bark

Meow

#Task 9

```
import numpy as np
```

```
a = np.arange(1, 11)
print("1d Array:", a)
```

```
zero = np.zeros((3, 3))
print("\nZero matrix:\n", zero)
```

```
identity = np.eye(3)
print("\nIdentity matrix:\n", identity)
```

```
rand = np.random.randint(1, 101, 5)
print("\nRandom numbers:", rand)
```

```
reshape = a.reshape(2, 5)
print("\nReshaped:\n", reshape)
```

```
sorted_des = np.sort(a)[::-1]
print("\nSorted in descending order:", sorted_des)
```

```
print()
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship,
1d Array: [ 1  2  3  4  5  6  7  8  9 10]
```

```
Zero matrix:
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
```

```
Identity matrix:
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

```
Random numbers: [ 1 70 78 77 63]
```

```
Reshaped:
[[ 1  2  3  4  5]
 [ 6  7  8  9 10]]
```

```
Sorted in descending order: [10  9  8  7  6  5  4  3  2  1]
```

```

# Task 10

import pandas as pd

data = {
    "Name": ["Subrahmanya", "Ravi", "Sita", "Ravi"],
    "Age": [20, 22, None, 22],
    "Marks": [85, 90, 88, 90]
}

df = pd.DataFrame(data)
df.to_csv("students.csv", index=False)
df = pd.read_csv("students.csv")

print(df.head())
print(df.describe())

df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Marks"] = df["Marks"].fillna(df["Marks"].median())

df = df.drop_duplicates()
df = df.rename(columns={"Marks": "Score"})

df.to_csv("cleaned_students.csv", index=False)
print(df)

```

```
===== RESTART: C:/Users/krsu/OneI
```

	Name	Age	Marks
0	Subrahmanya	20.0	85
1	Ravi	22.0	90
2	Sita	NaN	88
3	Ravi	22.0	90

	Age	Marks
count	3.000000	4.000000
mean	21.333333	88.250000
std	1.154701	2.362908
min	20.000000	85.000000
25%	21.000000	87.250000
50%	22.000000	89.000000
75%	22.000000	90.000000
max	22.000000	90.000000

	Name	Age	Score
0	Subrahmanya	20.000000	85
1	Ravi	22.000000	90
2	Sita	21.333333	88

```

# Task 11
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 20, 25, 30, 40]

plt.plot(x, y)
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Line Plot")
plt.show()

plt.bar(x, y)
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Bar Chart")
plt.show()

plt.pie(y, labels=x, autopct='%1.1f%%')
plt.title("Pie Chart")
plt.show()

plt.hist(y)
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.title("Histogram")
plt.show()

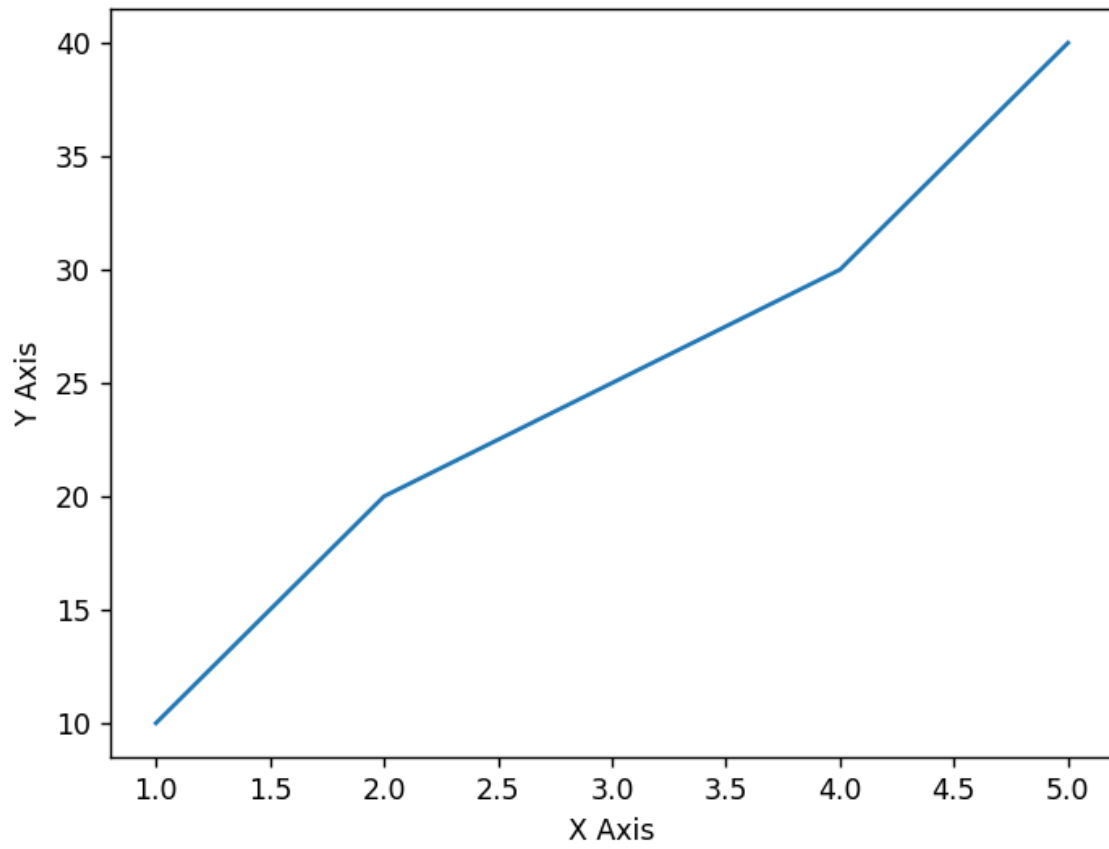
plt.scatter(x, y)
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Scatter Plot")
plt.show()

plt.subplot(2, 1, 1)
plt.plot(x, y)
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Subplot Line")
plt.subplot(2, 1, 2)
plt.bar(x, y)
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Subplot Bar")

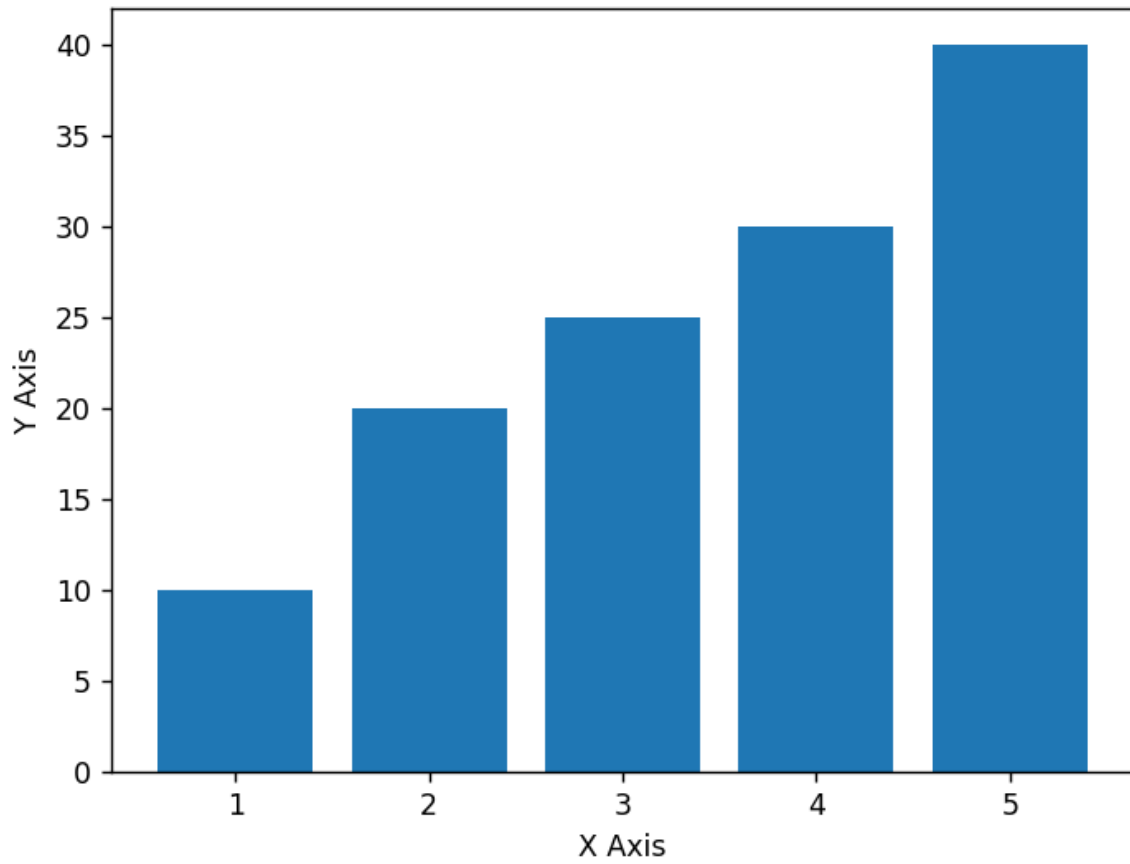
plt.tight_layout()
plt.show()

```

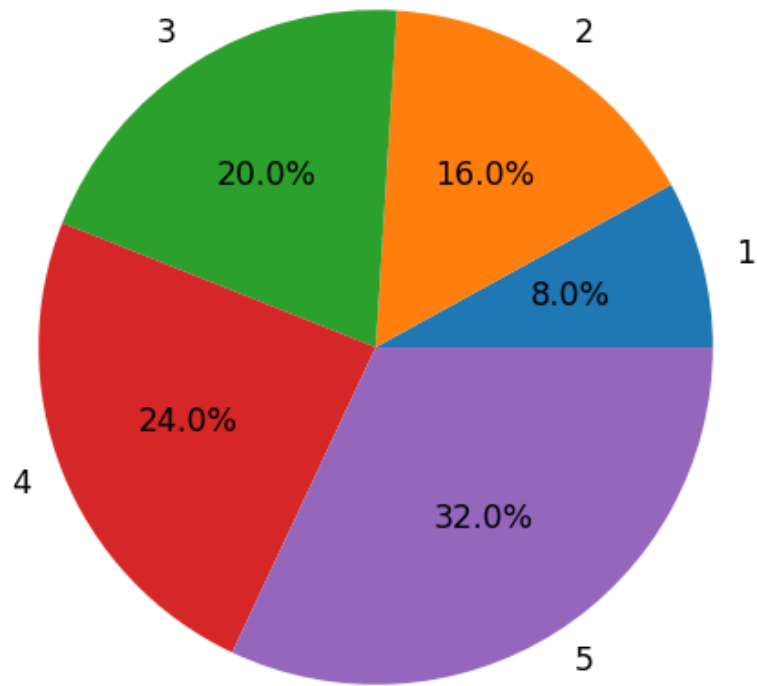
Line Plot

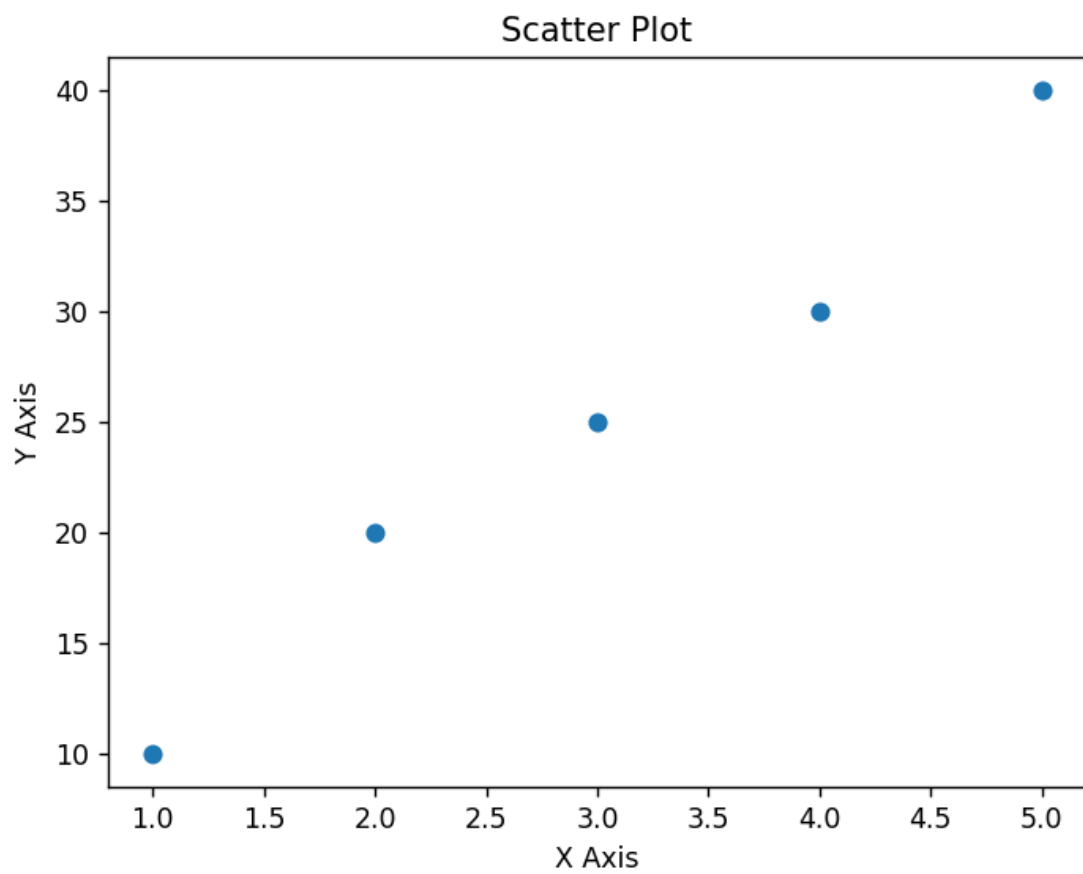
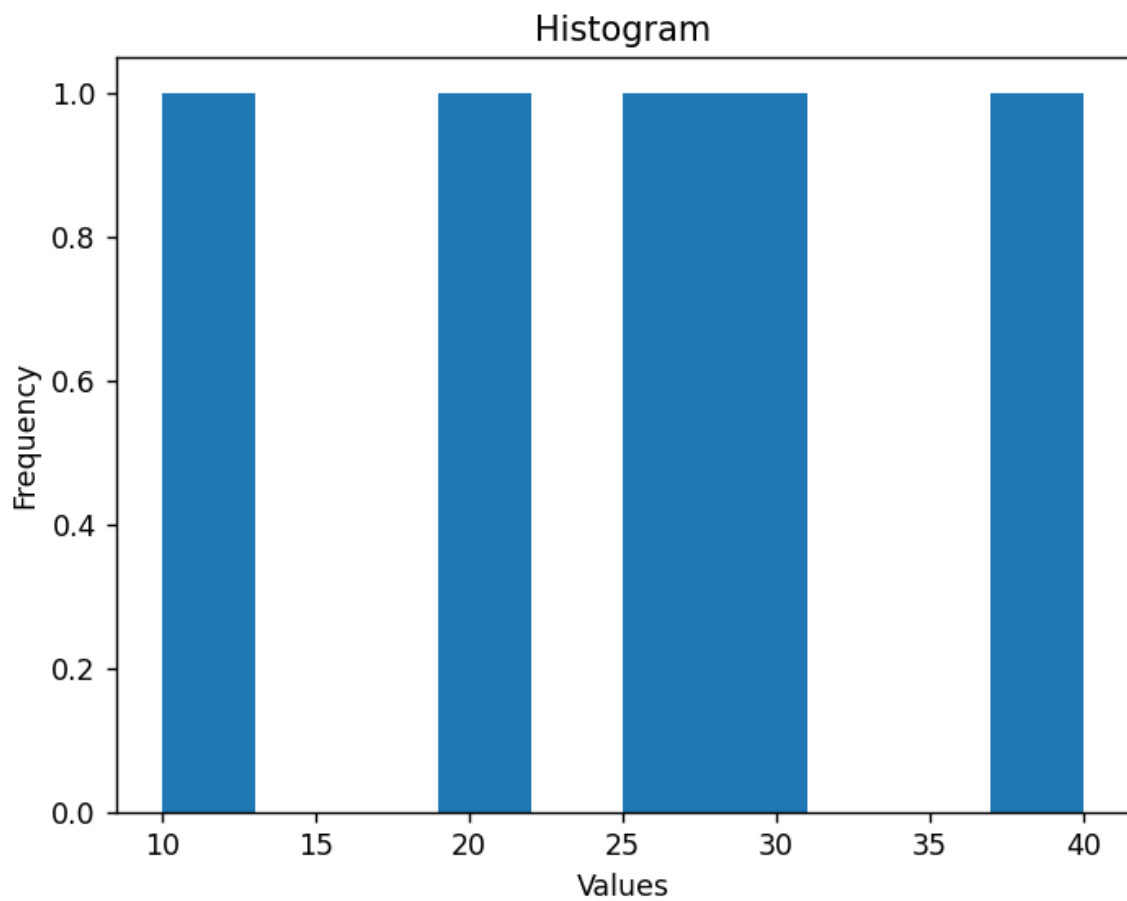


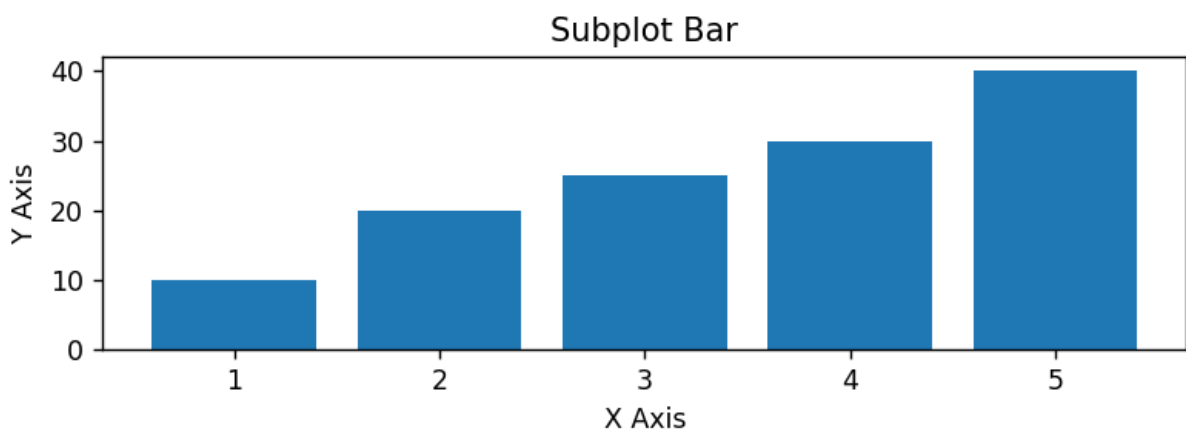
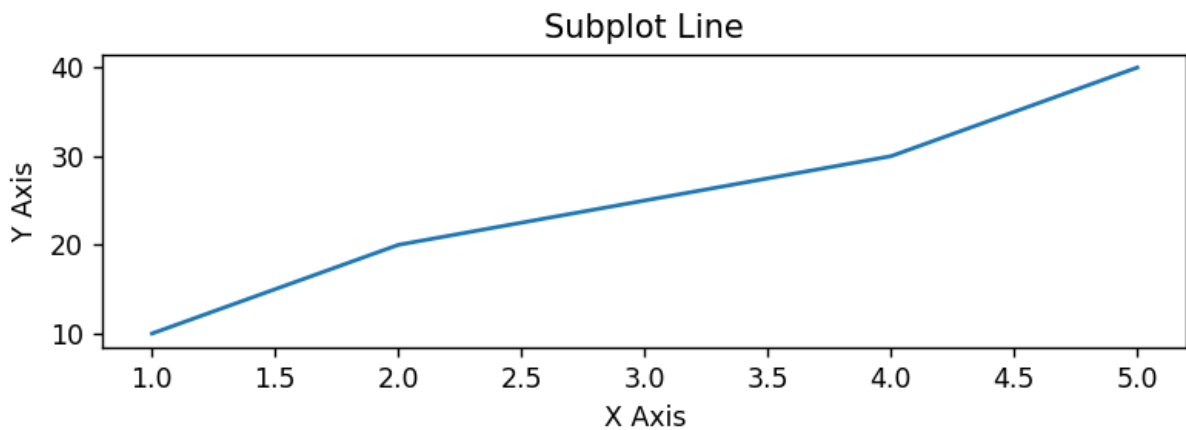
Bar Chart



Pie Chart







```
# Task 12
from PIL import Image, ImageFilter

img = Image.open("car.jpg")
img.show()

resized = img.resize((300, 300))
resized.show()

gray = img.convert("L")
gray.show()

rotated = img.rotate(90)
rotated.show()

cropped = img.crop((50, 50, 250, 250))
cropped.show()

blurred = img.filter(ImageFilter.BLUR)
blurred.show()

blurred.save("blurred_image.jpg")
img.save("converted_image.png")
```

